

# Ethan C. Morgan

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## ML RESEARCH

### Verifier-Backed Coding RL with GRPO (CS224N) – Qwen2.5-Coder-7B, TRL

Mar. 2026

- Engineered a high-throughput verifier-backed coding-RL environment by replacing per-rollout SWE-smith Docker builds with prewarmed local Rust workspaces, cached dependencies, and parallel compiler/test execution; cut median G=8 reward evaluation to 4.1 seconds.
- Ablated output formats, context lengths, group size, and reward shaping; diagnosed a silent all-zero toolchain failure and filtered unsound tasks, raising GRPO test-pass rate over SFT from 20.3% to 32.9% across 419 held-out tasks.
- Audited the complete evaluation set for reward hacking; 99.2% of test-passing outputs were legitimate code fixes.

### Visual SDPO for Computer Use (CS224R) – Qwen3-VL-8B, verl, vLLM

Jul. 2026

- Extended Self-Distillation Policy Optimization (SDPO) to visual GUI grounding in verl by adding privileged screenshot-conditioned teacher re-scoring; trained Qwen3-VL-8B across 8 B200 GPUs and reached a project-best 94.1% held-out ScreenSpot accuracy.
- Ran SFT, GRPO, SDPO, context-length, and optimization ablations while resolving OOMs and a vLLM CUDA-graph failure; built a teacher-probe that predicted opposite SDPO outcomes on ScreenSpot and ScreenSpot-Pro ( $p < 0.01$ ).

## WORK EXPERIENCE

### Amazon AGI Autonomy, San Francisco CA

Member of Technical Staff

Apr. – Jul. 2026

- Co-built an end-to-end synthetic-data factory for real-time computer use—task generation, Windows-VM execution and recording, LLM and deterministic verification, and SFT export—supplying training with 200 GB/hour of verified trajectories.
- Cut SFT data-preparation time 40% by profiling the data path and replacing full-video downloads with byte-range retrieval before parsing and training.

Software Development Engineer II

Dec. 2024 – Apr. 2026

- Founding engineer who launched Amazon Nova Act, AWS's computer-use-agent service; built core APIs, backend infrastructure, and Amazon security integration on a six-engineer team.
- Productionized the 76B model-inference pipeline for Amazon Nova Act by scaling SageMaker-backed clusters, automating releases, and hardening paths for public launch.
- Built observer-model guardrails on a separate inference stack and improved real-time throughput 8x through batching.
- Took the research prototype through external-preview, enterprise, and AWS-service launches for browser workflows including web scraping, visual validation, Q&A, and form completion.

### AWS Cryptography, Credential Distribution Systems, Seattle WA

Software Development Engineer II

Jul. – Dec. 2024

- Led the design and production rollout of deterministic re-sharding for a Tier-0 cryptography service supporting nearly all Amazon systems, removing a fixed 80-shard capacity ceiling and increasing capacity 10x.
- Resolved the top customer pain point, reducing reach outs by 43% (15 per week) and saving 19 hours per week of repeated support work.
- Mentored 2 junior engineers and 1 intern through design, implementation, and delivery of high-quality production systems.

### AWS Cryptography, Credential Distribution Systems, Seattle WA

Software Development Engineer I

Mar. 2023 – Jun. 2024

- Automated customer onboarding for 5,000+ customers for our internal product with a React/Java website.
- Delivered two AWS-wide wins: refactored Java logging to save \$10M/year and cut Python monitoring runtime by 95% via profiling and parallelism; each was showcased to 2,000+ leaders.
- Built a Python Assistant Chatbot (AWS Bedrock + RAG/Agents) for answering customer questions, winning "Best Project" in an org-wide hackathon.

## EDUCATION

### Stanford University

College credit: CS224R (Deep RL); CS224N (Deep NLP)

Jan. – Jul. 2026

### University of Michigan, Ann Arbor, MI

Bachelor of Science in Computer Science

Dec. 2022

## SKILLS

**Programming:** Python, Java, Rust, TypeScript/React, C/C++, SQL

**ML / Systems:** Hugging Face, TRL, verl, vLLM, FSDP, CUDA, SageMaker, AWS, CloudWatch, CloudFormation, Linux, Git